Selcuk Kilinc *Postdoctoral Researcher*

Catskill 143 at School of Education | University at Albany   skilinc@albany.edu   +1 979 721 3467   Male   LinkedIn
ResearchGate   Google Scholar   0000-0001-8846-7243   Personal Website

Postdoctoral researcher in science and STEM education specializing in AI-supported teaching and learning, with a primary research focus on AI literacy, generative AI, and mixed-methods studies of how AI tools support inquiry, reflection, and pedagogical decision-making.

EDUCATION

Doctor of Philosophy / Science Education 02/2018 – 07/2023 | Turkey
Middle East Technical University
- **Dissertation:** Real-Time Air Quality Monitoring as A Catalyst for the Growth of Pre-Service Science Teachers in Citizen Science

Master of Science / Software Engineering 09/2020 – 07/2022 | Turkey
Middle East Technical University
- ***Graduation Project:*** SmartCal: Innovating Mobile Tech for Real-Time Calorie Tracking and Dietary Guidance

Master of Science / Secondary Science and Mathematics Education 01/2015 – 01/2018 | Turkey
Middle East Technical University
- **Thesis:** Exploring of STEM Readiness of a Faculty of Education in Turkey

Integrated Bachelor's and Non-Thesis Master's Chemistry Education 09/2007 – 06/2014 | Turkey
Middle East Technical University
- ***Graduation Project:*** Goal Orientations and Task Values of High School Students

PROFESSIONAL EXPERIENCE

University at Albany, State University of New York 01/2026 – Present | NY, USA
Postdoctoral Research Scholar
- Conducting research on artificial intelligence in education, with a focus on AI-supported teaching and learning in higher education.
- Examining how generative AI tools are conceptualized and used in educational contexts, including implications for learning and assessment.
- Supporting scholarly publications, grant-related activities, and interdisciplinary collaborations in AI in education, learning sciences, and edtech.

Texas A&M University 01/2025 – 12/2025 | TX, USA
Postdoctoral Research Associate
- Co-designed and implemented an AI-enhanced Ag–STEM program across two rural school districts, reaching 80+ middle school students.
- Integrated AI chatbot tutors and Arduino-based IoT sensors to scaffold computational thinking and inquiry in classroom instruction.
- Led teacher professional development and classroom support, contributing to grant proposals, invited lectures, and peer-reviewed publications.

Middle East Technical University Faculty of Education

Research & Teaching Assistant 09/2015 – 12/2024 | Turkey
- Conducted research, published papers, and presented at international conferences in science education and educational technology
- Designed and taught undergraduate and graduate courses in instructional technology, research methods, and science pedagogy
- Developed AI- and IoT-enhanced learning environments and mentored students in qualitative, quantitative, and mixed-methods research.

Coordinator for Research Assistants
- Led 20+ assistants, strategically aligning their responsibilities with faculty priorities and coordinating across departmental initiatives.
- Implemented efficient scheduling and workload distribution systems, ensuring the timely completion of research and instructional duties.

Distance Education Coordinator
- Helped lead the faculty's transition to online education during COVID-19, supporting 40+ faculty and 200+ students across the faculty.
- Collaborated with instructors to develop interactive online courses and used learning analytics to inform improvements.

Technology Coordinator
- Provided technical support and training on educational technologies for 40+ faculty members and staff.
- Supported the redesign and maintenance of the department website, improving navigation and access to key information.

Jale Tezer High School 09/2014 – 08/2015 | Turkey
Chemistry Teacher
- Planned and delivered curriculum-aligned chemistry instruction for 300+ high school students with hands-on experiments.
- Designed and implemented online educational materials to support differentiated instruction.

HONORS & AWARDS

TUBITAK 2219 International Postdoctoral Research Fellowship Program 2025
The Scientific and Technological Research Council of Turkey
- Awarded a competitive postdoctoral research fellowship ($23,400) to research AI-enhanced STEM education at Texas A&M University, USA.

TUBITAK 2224-A – International Conference Travel Grant 2025
Overseas Scientific Meetings Participation Support Program
- Received approximately $1,700 in travel and registration support to present research at the NARST 2025 Annual International Conference.

Dean's Honor List Student 2013
Middle East Technical University (METU)
- Placed on the university honor list twice (January 2013 and July 2013) in recognition of high academic achievement during BS studies.

PUBLICATIONS

Peer-Reviewed Journal Articles

- Evkaya, O., **Kilinc, S.**, & Kizilates, S. (2026). Cross-national perceptions of generative AI in higher education: A comparative study of university students in the UK and Turkey. *International Journal of Information and Education Technology, 16*(3), 640–647. https://doi.org/10.18178/ijiet.2026.16.3.2536
- Aldemir T., **Kilinc S.**, Bicer A., Grant P., Davis, T. J. & Sweany, N. W. (2025). Intelligent-TPACK in Practice: Design and Evidence from a Three-Week Teacher Preparation Module. *Computers and Education Open*, 9, 100306. https://doi.org/10.1016/j.caeo.2025.100306
- Aldemir T., Bicer A., **Kilinc S.**, Moon J., & Kwok, M. (2025). Exploring Emergent AI-TPACK Competencies in a Two-Week AI Literacy Module for Preservice Teachers. *Teaching and Teacher Education*, 168, 105231. https://doi.org/10.1016/j.tate.2025.105231
- **Kilinc, S.,** Geban, O. & Ozturk, G. (2025). STEM in Teacher Education: Readiness, Challenges, and Alignment with Global Frameworks. *Ankara University Journal of Faculty of Educational Sciences (JFES)*, 58(3), 981-1032.. https://doi.org/10.30964/auebfd.1567788
- **Kilinc, S.** (2024). Comprehensive AI assessment framework: Enhancing educational evaluation with ethical AI integration. *Journal of Educational Technology and Online Learning*, *7*(4-ICETOL 2024 Special Issue), 521-540. https://doi.org/10.31681/jetol.1492695
- **Kilinc, S.** (2023). Embracing the Future of Distance Science Education: Opportunities and Challenges of ChatGPT Integration. *Asian Journal of Distance Education*, *18*(1), 205-237. https://doi.org/10.5281/zenodo.7857396
- Uzuntiryaki-Kondakci, E., Tuysuz, M., Sarici, E., Soysal, C., & **Kilinc, S.** (2021). The role of the argumentation-based laboratory on the development of pre-service chemistry teachers' argumentation skills. *International Journal of Science Education*, *43*(1), 30-55. https://doi.org/10.1080/09500693.2020.1846226

Peer-Reviewed Books Chapters

- **Kilinc, S.** (2025). Personalizing Education in the AI Era: The Comprehensive Impact of Customized Chatbots Across Educational Domains. In *Artificial Intelligence and Human Agency in Education: Volume Two: AI for Equity, Well-Being, and Innovation in Teaching and Learning*. https://doi.org/10.1007/978-981-96-9251-4_2

Peer-Reviewed Proceedings

- **Kilinc, S.,** Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2025). AI Chatbot Coaching for Elevating Student Research. In *Proceedings of the 2025 ACM Conference on International Computing Education Research V. 2*. https://doi.org/10.1145/3702653.3744315
- **Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & So, D. (2026). Planners and tinkerers: A comparative case study of K-12 students' engagement with data practices in an Ag-STEM. In *Proceedings of the International Conference of the Learning Sciences – ICLS 2026*, pp. xx–xx. International Society of the Learning Sciences. *(accepted, in-press)*
- Sabanwar, V., Aldemir, T., **Kilinc, S.**, Bicer, A., & Song, D. (2026). How middle school students engage in computational literacies in an agricultural class. In *Proceedings of the International Conference of the Learning Sciences – ICLS 2026*, pp. xx–xx. International Society of the Learning Sciences. *(accepted, in-press)*
- Aldemir, T., **Kilinc, S.**, Sabanwar, V., Bicer, A., & Song, D. (2026). Disciplinary computational thinking in AI-mediated problem framing: A rural Ag-STEM context. In *Proceedings of the International Conference of the Learning Sciences – ICLS 2026*, pp. xx–xx. International Society of the Learning Sciences. *(accepted, in-press)*

Manuscripts Under Review

- **Kilinc S.,** Aldemir T., Misiejuk K., Kaliisa R., Bicer A., Sabanwar, V., Song, D., Yalvac B. (2025). A Mixed-Methods Analysis of AI Scaffolding Patterns and Student Inquiry Profiles in a Middle-School Agriculture-STEM Classroom. *Computers & Education: Artificial Intelligence.*
- Aldemir T., Bicer A., **Kilinc S.**, Moon J. & Kwok, M. (2025). Challenges, Solutions, and PD Needs for Integrating AI: Insights from a Two-Week AI Literacy Module with Preservice Teachers. *Action in Teacher Education.*

PRESENTATIONS

Peer-Reviewed Conference Presentations

- **Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & So, D. (2026). *Planners and tinkerers: A comparative case study of K-12 students' engagement with data practices in an Ag-STEM.* Accepted for presentation as a Long Paper at the International Conference of the Learning Sciences (ICLS) 2026.
- Sabanwar, V., Aldemir, T., **Kilinc, S.**, Bicer, A., & Song, D. (2026). *How middle school students engage in computational literacies in an agricultural class.* Accepted for presentation as a Long Paper at the International Conference of the Learning Sciences (ICLS) 2026.
- Aldemir, T., **Kilinc, S.**, Sabanwar, V., Bicer, A., & Song, D. (2026). *Disciplinary computational thinking in AI-mediated problem framing: A rural Ag-STEM context.* Accepted for presentation as a Poster at the International Conference of the Learning Sciences (ICLS) 2026.
- **Kilinc, S.**, Yilmazoglu, E., İncecay, İ., & Kondakci, E. (2026). *From Tool to Teammate: Reimagining Student-AI Collaboration in Engineering Design-Based STEM.* Accepted for presentation at the National Association for Research in Science Teaching (NARST) Annual International Conference, Seattle, WA, USA.
- Kondakci, E., İncecay, İ., Yilmazoglu, E., & **Kilinc, S.** (2026). *Enhancing High School Students' Self-Regulation with an AI-supported Engineering Design Process.* Accepted for presentation at the National Association for Research in Science Teaching (NARST) Annual International Conference, Seattle, WA, USA.
- **Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2026). *The AI as Cognitive Mentor: Scaffolding Student Scientific Inquiry Through Dialogue.* Accepted for presentation at the American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA, USA.
- **Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2026). *Computational Thinking and Participation in Rural Ag STEM: A Comparative Case Study of Two Middle School Teams.* Accepted for presentation at the American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA, USA.
- Sabanwar, V., Aldemir, T., **Kilinc, S.**, Bicer, A., & Song, D. (2026). *Teacher Conceptualization of Student Challenges and Solutions for CT in A Non-CS Classroom.* Accepted for poster presentation at the American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA, USA.
- Evkaya, O., **Kilinc, S**, Kizilates S. (2025). *Can an AI-Powered Game-based Presentation Platform Be Used to Improve Students' Public Speaking Skills?.* Presented at the International Conference on Education and Artificial Intelligence Technologies (EAIT) Convention, Manchester, UK.
- **Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2025). *Enhancing Research Questions and Hypothesis Development Through AI Chatbot Assistance in Rural Agricultural Education*. Presented at the Association of Educational Communications & Technology (AECT) International Convention, Las Vegas, NV, USA.
- Sabanwar, V., Aldemir, T., **Kilinc, S.**, Bicer, A., & Song, D. (2025). *Exploring Computational Thinking skills in the era of GenAI in K–12 education.* Presented at the Association of Educational Communications & Technology (AECT) International Convention, Las Vegas, NV, USA.

- **Kilinc, S.**, Aldemir, T., Sabanwar, V., Bicer, A., & Song, D. (2025). *AI Chatbot Coaching for Elevating Student Research.* Poster presented at the ACM Conference on International Computing Education Research (ICER), Charlottesville, VA, USA.
- **Kilinc, S.** & Ozturk, G. (2025). *Innovative Hybrid Science Education: Integrating Citizen Science & Digital Learning for Future-Ready Teachers.* Presented at the National Association for Research in Science Teaching (NARST) Annual International Conference, National Harbor, MD, USA.
- **Kilinc, S.**, & Ozturk, G. (2024). *Evaluation of the Effect of Citizen Science Projects on the Scientific Competences of Pre-Service Teachers.* Presented at the National Science and Mathematics Education Congress, Edirne, Turkey.
- **Kilinc, S.** (2024). *Shifting Paradigms: Ethical Approaches and Integrity Guidelines for AI-Assisted Assessments in Learning.* Presented at the International Conference on Educational Technology and Online Learning (ICETOL), Eskisehir, Turkey.
- **Kilinc, S.**, Geban, O., & Ozturk, G. (2018). *Exploring Instructors' Perceptions of A Faculty Of Education In Turkey About STEM Education*. Presented at the Annual European Conference on Educational Research (ECER), Bolzano, Italy.
- **Kilinc, S.**, Sarici, E., Soysal, C., & Kondakci, E. (2017). *Contribution of the argumentation-based laboratory to pre-service chemistry teachers' microscopic explanations of chemistry concepts.* Presented at the National Association for Research in Science Teaching (NARST) Annual International Conference, San Antonio, TX, USA.
- Soysal, C., Sarici, E., **Kilinc, S.**, & Kondakci, E. (2017). *Development of Pre-service Chemistry Teachers' Argumentation Skills in Implementing Science Writing Heuristic at Chemistry Laboratory Subject/Problem*. Presented at the National Association for Research in Science Teaching (NARST) Annual International Conference, San Antonio, TX, USA.
- Ekiz Kiran, B., Kutucu, E. S., **Kilinc, S.**, Soysal, C., & Boz, Y. (2016). *Pre-Service Chemistry Teachers' Level of Explaining Daily Life Events Using Their Chemistry Knowledge.* Presented at the National Science and Mathematics Education Congress, Trabzon, Turkey.

Invited Lectures & Workshops

- From Questions to Insights: Demonstrating Qualitative Design in Action (2025). *Guest lecture for EDCI 690, Department of Teaching, Learning & Culture, Texas A&M University, USA.*
- Citizen-Science + IoT: Pathways to Authentic STEM & Future-Ready Teachers (2025). *Guest lecture for ENGR/SCIED 497F, Department of Teaching, Learning & Culture, Texas A&M University, USA.*
- Generative AI and Custom GPT Usage in STEM Projects (2025). *Invited lecture, TED Ankara College (Online), Ankara, Turkey.*
- Citizen Science Projects in Authentic Research (2023). *Workshop on Climate Change and Environment, Ankara, Turkey.*
- Distance Education and LMS Integration into Teaching (2020 & 2021). *Workshop, Middle East Technical University, Ankara, Turkey.*

PROJECTS

Artificial Intelligence Mentorship for Rural Education 01/2025 – 01/2026 | Texas, USA
Educational Innovation Project
- Co-designed and implemented a 16-week AI-enhanced agriculture-STEM curriculum for 80+ middle school students, emphasizing computational thinking and data-driven environmental investigation.
- Integrated Arduino-based IoT sensors and a custom AI chatbot tutor to scaffold inquiry, troubleshooting, and analysis of authentic environmental data.
- Led curriculum co-design and teacher professional development, supporting implementation across Caldwell and Brenham Junior High Schools and generating data for conference presentations and journal manuscripts.

Permaculture Chatbot & Engineering-Design Mentorship 10/2024 – Present | Turkey
Educational Design Project
- Developed a chatbot to support permaculture learning and engineering-design mentorship, informed by a design-based research framework.
- Implemented an instructional intervention with ~70 students, including a two-day introductory unit and a subsequent two-month project phase where students used a custom chatbot to develop and iterate on permaculture design projects.
- Iteratively refined the system based on user feedback, interaction data, and emerging learner needs.
- Generated design insights into how AI-supported mentorship can scaffold ecological design thinking and problem-solving.
- Currently implementing the second phase of the project, integrating lessons learned from the initial deployment to improve functionality and pedagogical alignment.

IoT-Enabled Citizen Science: Real-Time Air Quality Monitoring 09/2022 – 02/2023 | Turkey
Dissertation Project
- Designed and built WHO-aligned Arduino-based mobile air-quality monitor ($PM_{2.5}$, PM_{10}, key gas pollutants) with real-time display and data log.
- Integrated the device into a science methods course with 12 preservice teachers as an authentic citizen-science project.
- Supported preservice teachers in posing research questions, collecting air-quality data across diverse campus locations, and analyzing and visualizing the datasets.
- Facilitated data-informed presentations and recommendations to university leadership and documented the project in a short field video.

Determining of Education Faculties' STEM Readiness 01/2017 – 12/2017 | Turkey
Scientific Research Project
- Assessed the STEM readiness of education faculty members and preservice teachers after STEM education was introduced into the curriculum.
- Collected survey and interview data to identify needs in instructional technology use, interdisciplinary collaboration, and perceptions of STEM.
- Produced a policy report that informed Turkish Higher Education Council professional-development and curriculum-reform efforts.

TEACHING EXPERIENCES

U.S. Teaching & Professional Development 01/2025 – 12/2025 | TX, USA
Texas A&M University & Caldwell Junior High School
- Co-designed and taught a 16-week AI-enhanced agriculture–STEM program for 80+ junior high students, using co-teaching cycles and independently led lessons to support students' development of computational thinking and scientific reasoning.
- Modeled inquiry- and project-based STEM instruction, providing real-time scaffolding as students used AI chatbots and Arduino-based IoT sensors to investigate authentic agricultural problems.
- Designed and facilitated ongoing, job-embedded professional development and coaching for partner teachers on integrating AI tools and sensor data into their daily lessons and assessment practices.
- Mentored a PhD student in research design, data collection, validity and reliability, and academic writing for publication.

University-Level Instruction & Mentoring 09/2015 – 12/2024 | Turkey

Middle East Technical University

- **Instructor & Course Designer (Undergraduate & Graduate Courses)**
  - *SSME518 & MSE329 - Instructional Tech & Material Development (Face-to-Face & Hybrid, BS & MS)*
    - Led both graduate and undergraduate courses from concept to execution, including syllabus design, project-based learning activities, and assessment frameworks aligned with inquiry-based STEM education.
    - Designed curricula focused on developing IoT-based environmental monitoring projects using Arduino, enabling students to collect, analyze, and interpret real-world environmental data.
    - Pioneered the integration of Generative AI into the course, empowering students to create innovative, adaptive learning content.
  - *MSE405 - Lab Applications in Science Education I (Face-to-Face, BS)*
    - Headed hands-on laboratory sessions for chemistry education, focusing on building students' critical thinking, experimental design, and scientific reasoning skills.
- **Teaching Assistant & Mentor (Graduate Courses)**
  - *MSE603, MSE602, MSE502 - Research Methods & Qualitative Research & Advanced Research (Face-to-Face & Online, MS & PhD)*
    - Provided mentorship to MS/PhD students in qualitative and quantitative educational research methods and research ethics.
    - Mentored undergraduate and graduate students in designing technology-enhanced science education projects, including AI-supported instructional tools and sensor-based inquiry activities.
- **Teaching Assistant & Mentor (Undergraduate Courses)**
  - *MSE409 & MSE410 - Practice Teaching I and II (Face-to-Face, BS)*
    - Supervised and mentored pre-service teachers during school-based practicum experiences, providing structured feedback on lesson planning, classroom instruction, micro-teaching sessions, and teaching portfolios to support reflective practice and professional growth.
  - *MSE310 & MSE411 & MSE305 - Secondary Science Teaching Methods I and II & Assessment in Science Education (Face-to-Face, BS)*
    - Guided pre-service teachers in applying inquiry-based teaching methods and instructional strategies to design effective science lessons.
    - Guided pre-service teachers in designing and implementing assessment strategies, including formative and summative assessment, rubric development, and evaluation of student learning.

Teacher Mentoring & Instructional Design 09/2019 – 12/2024 | Turkey

Bestepe College & TED Ankara College

- Coached IB science teachers through year-long professional development cycles, including classroom observations, feedback conferences, and collaborative planning.
- Facilitated workshops on inquiry-based and interdisciplinary curriculum design, student-centered assessment, and the use of digital and AI tools to support K–12 science learning.
- Designed and delivered interactive sessions for students and teachers on the ethical and productive use of Generative AI to foster digital literacy.

K-12 Classroom & Intern Experience 09/2013 – 08/2015 | Turkey

Jale Tezer High School & Ankara Anatolian High School & METU High School

- Designed and delivered curriculum-aligned chemistry lessons for high school students, integrating hands-on laboratory experiments and digital resources to support diverse learners.
- Completed supervised teaching practicums in multiple high schools, collaborating with mentor teachers on lesson planning, lab management, and providing one-on-one academic support to students.

PROFESSIONAL SERVICE

Journal Reviewer

- Computers & Education Open
- Teaching and Teacher Education
- Frontiers in Education
- Education Innovations: Systems and Future Learning (EISFL)
- AIS Transactions on Human-Computer Interaction
- International Journal of Science, Technology and Society
- Acta Infologica (ACIN)
- Turkish Journal of Educational Sciences

Conference Program Committee

- The International Society of the Learning Sciences (ISLS) Annual Meeting 2025
- National Science and Mathematics Education Congress 2024

Conference Reviewer

- Association for Educational Communications and Technology International Convention 2026
- International Conference on Artificial Intelligence in Education 2026
- International Conference of the Learning Sciences 2026
- International Conference on Computer-Supported Collaborative Learning Annual Meeting 2026
- Association for Educational Communications and Technology Online Conference 2026
- National Association for Research in Science Teaching Annual International Conference 2026
- American Educational Research Association Annual Meeting 2026
- National Association for Research in Science Teaching Annual International Conference 2025
- National Science and Mathematics Education Congress 2024

CERTIFICATES

AI & Programming & Engineering

- Advanced AI App Design for Educators (PLC 201), Playlab.ai, 2025
- AI for Education Summit, AI for Education, 2024
- Career Essentials in Generative AI, Machine Learning, and Ethics, Microsoft & LinkedIn, 2023
- Python 3 Programming Specialization, University of Michigan (Coursera), 2022
- Complete Python Bootcamp: From Zero to Hero, Udemy, 2021
- Information Technologies Certificate (1-year program), METU CEC, 2021

Research Methodologies

- Instructional Design Foundations and Applications, University of Illinois Urbana–Champaign, 2025
- Data Science Methodology, IBM (Coursera – Alex Aklson & Polong Lin), 2022
- Qualitative Research, University of California, 2021
- Understanding Clinical Research: Behind the Statistics, Univ. of Cape Town, 2020
- Computer-Assisted Qualitative Data Analysis (MAXQDA certified training), 2019
- Methods & Statistics in Social Sciences Specialization, Univ. of Amsterdam, 2019

Leadership and Management

- Project Management & Career Development Specialization, University of California (Coursera), 2025
- Team & Project Management (32-week School of Leadership program), 2021
- Mentoring Program (8-month, Association for Coaching–accredited), 2019
- Creative Drama Leadership (1-year certificate), Contemporary Drama Association, 2014

SKILLS

Science Education & Learning Sciences
- Technology-enhanced K–12 and higher education science curricula design grounded in inquiry, modeling, and engineering design.
- Teacher professional development in science education and AI/EdTech integration, including workshop design, co-teaching, and ongoing coaching.
- Translating learning sciences and STEM education research into scalable classroom-ready tools, assessments, and PD resources.

Research Methodologies
- Mixed-methods research on technology-enhanced STEM and AI-in-education contexts.
- Qualitative approaches, including semi-structured interviews, focus groups, classroom observations, document and artifact analysis, and thematic/content analysis (MAXQDA).
- Quantitative approaches, including survey design and validation, scale development, experimental and quasi-experimental designs, and inferential statistics (t-tests, ANOVA, regression), plus instrument development for questionnaires, rubrics, and performance assessments.
- Learning analytics and multimodal data analysis, including log/clickstream and time-on-task data, video-based classroom interaction coding, and Epistemic Network Analysis (ENA).

AI & Instructional Technology & Software Engineering
- Design and evaluation of online, hybrid, and blended courses using instructional design models, TPACK, and AI-TPACK frameworks.
- Development of AI-driven tutoring systems, chatbots, and adaptive learning environments, with attention to ethics, feedback, and learner support.
- IoT-based learning environments using Arduino sensor networks and real-time data dashboards to support inquiry and data literacy.
- Programming for educational research and data systems (Python, basic Java) and rapid prototyping of research tools and pipelines.
- LMS and educational platforms (Canvas, Moodle, Google Classroom, etc.) for course design, analytics, and integration of external tools.

Collaboration & Leadership
- Mentoring K–12 teachers and preservice teachers in STEM, AI literacy, assessment, and classroom implementation of new tools.
- Coordinating interdisciplinary teams of educators, technologists, designers, and researchers across schools and universities.
- Competitive grant development, reporting, and multi-site project coordination with school districts and community partners.
- Designing & facilitating professional development workshops and co-design sessions in STEM and AI in education.

MEMBERSHIP IN PROFESSIONAL ASSOCIATIONS

- The International Society of the Learning Sciences (ISLS)
- National Association for Research in Science Teaching (NARST)
- American Educational Research Association (AERA)
- Association for Computing Machinery (ACM)
- Association for Educational Communications and Technology (AECT)

COMMUNITY ENGAGEMENTS

ANKA Youth Association 2020 – present
Society Member
- Organized international youth projects under Erasmus+ (e.g., “EthiTech: Fostering Responsible Digital Citizenship in Youth” in Romania), recruiting participants and promoting digital literacy and intercultural dialogue.
- Coordinated Speaking Clubs (e.g., “AI & The Future”, “Is Money Key to Happiness”), facilitating discussions on AI, ethics, and societal issues.
- Supported hybrid internship programs (e.g., “Unlock Your Potential” with Jugendvision e.V., Germany), connecting Turkish youth to global opportunities in project management and digital media.
- Contributed to the “#DijitalKöprüler: Strengthening Intercultural Dialogue through Responsible Digital Citizenship” event (Antalya, Turkey), featuring workshops on digital ethics, online security, and intercultural exchange—aligning with the themes of my dissertation.

Turkish Intelligence Foundation 2018 – present
Volunteer Staff
- Contributed to the annual “Intelligence and Talent Congress” (2013–Present), assisting with logistics and facilitating sessions on intelligence, talent development, and educational innovation at METU.
- Supported Turkey Intelligence Foundations Meetings (animation science and healthy nutrition), engaging educators and students.
- Assisted with session planning, breakout workshops, and expert panel logistics addressing topics such as gifted education, innovative testing methods, and the role of social communication in talent development.
- Supported the “Inter-School Intelligence Games Championship” (2019-present), coordinating nationwide events to enhance students’ problem-solving skills.

Darüşşafaka Society 2017 – present
Community Volunteer
- Participated in fundraising campaigns with companies and individuals to secure donations for underprivileged students’ education.
- Served as a mentor in the society’s mentorship program, guiding graduates in career planning and academic growth by one-on-one support.
- Facilitated student-professional networking events, connecting students with industry experts to promote educational equity.

METU Alumni Association 2016 – present
Member
- Contributed to various activities—including social, cultural, and professional development events—and actively participated in committees (e.g., Event, Communication, and Disaster & Emergency Assistance Committees).
- Coordinated and facilitated mentorship programs for numerous university students, providing career guidance and networking opportunities.
- Played a significant role in the February 2023 Turkey earthquake relief efforts by organizing donation drives, coordinating packaging, and ensuring efficient aid distribution to affected communities.
- Collaborated with cross-functional teams to enhance volunteer engagement and streamline logistical operations for big events.

The Educational Volunteers Foundation of Turkey 2013 – present
Education Volunteer
- Actively contributed to various fundraising events and donation campaigns that support underprivileged children’s educational opportunities.
- Supported developing and disseminating engaging educational content through TEGV Akademi for children, parents, volunteers, and teachers.
- Participated in aid collection, packaging, and distribution activities, particularly during the February 2023 earthquake.

LANGUAGES

Turkish ●●●●●
Native Language

German ●●●○○
Basic German skills with continual improvement

English ●●●●●
Advanced English is demonstrated through academic pursuits, including earning degrees, publishing research, and presenting at international conferences.